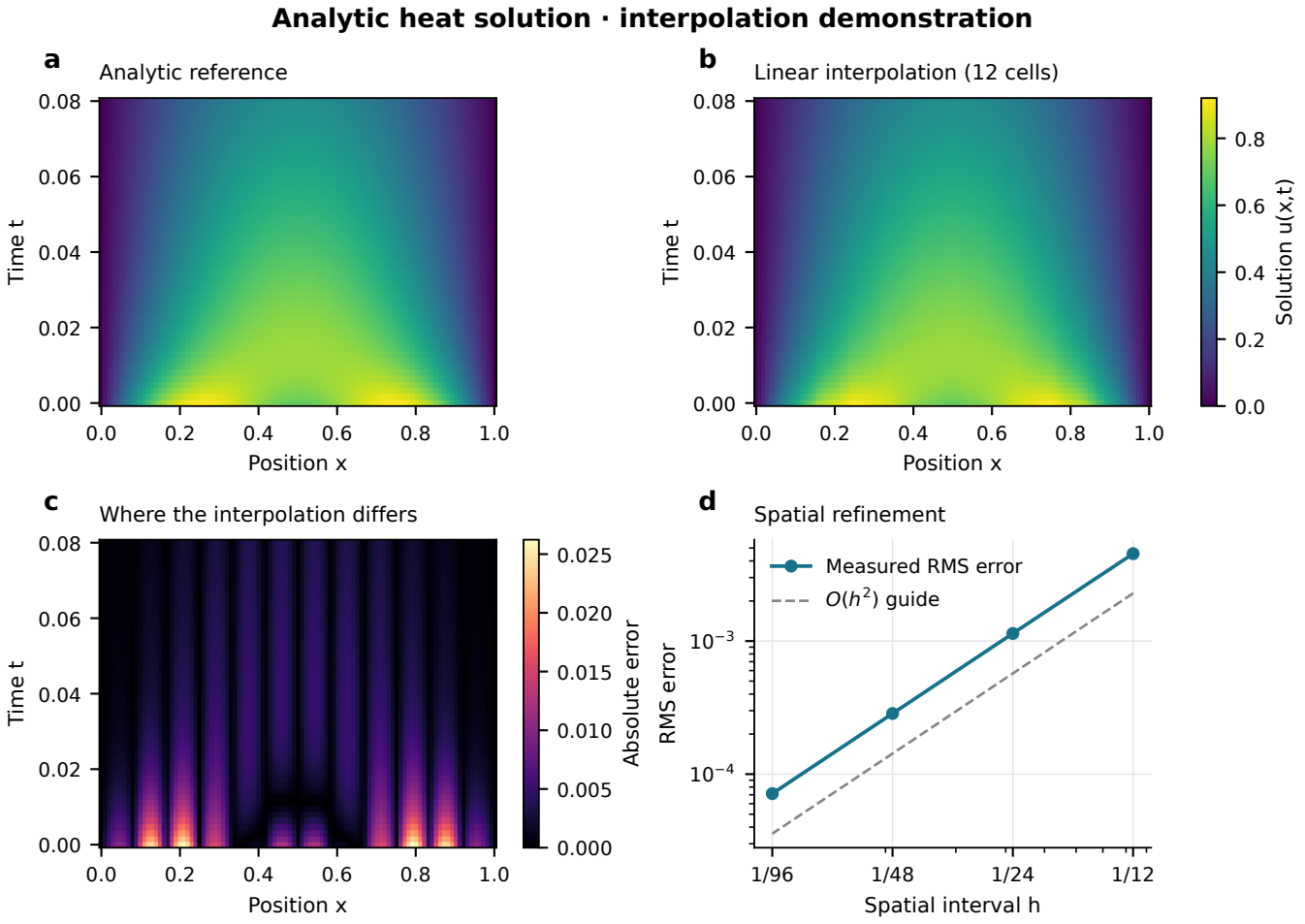

Analytic heat solution · interpolation demonstration
a
Analytic reference
b
Linear interpolation (12 cells)
c
Where the interpolation differs
d
Spatial refinement
Time t
Position x
Solution u(x,t)
Absolute error
RMS error
Spatial interval h
Measured RMS error
$O(h^2)$ guide
0.00
0.02
0.04
0.06
0.08
0.0
0.2
0.4
0.6
0.8
1.0
0.000
0.005
0.010
0.015
0.020
0.025
10^{-3}
10^{-4}
1/96
1/48
1/24
1/12